INCH-POUND

MIL-DTL-17/28C
w/AMENDMENT 1
13 April 2010

SUPERSEDING
MIL-C-17/28C
18 July 1985

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/28-RG058

Inactive for new design after 13 August 1993. For new design use MIL-C-17/183.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

NOTE: This cable uses PVC material and is not to be used in enclosed environments or shipboard applications. The replacements that are to be used in enclosed areas or shipboard applications are referenced in the following table.

The Air Force has restricted use of PVC in aerospace and ground support applications.

Cables listed on the current QPL may continue to be manufactured and supplied for existing enclosed applications only for a period not to exceed 3 years from the date of this specification.

TABLE I. Cross-reference data.

Current Part or Identifying Number (PIN)	Replacement PIN
M17/28-RG058	M17/183-00001

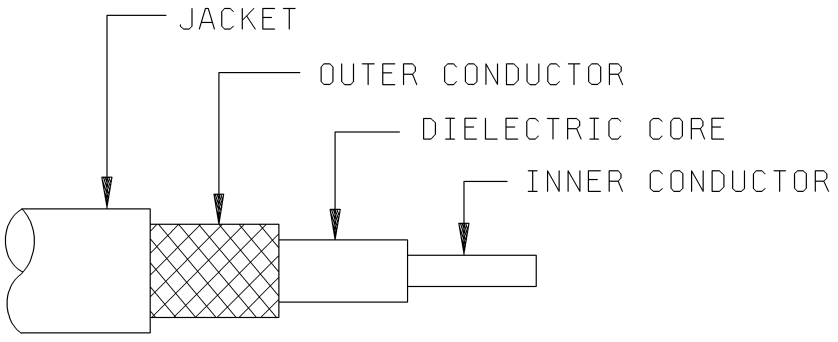


FIGURE 1. Configuration.

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TABLE II. Description.

Components	Construction details
Inner conductor	Nineteen strands of tinned, copper wire, .each strand 0.0072 inch diameter. Overall diameter: 0.0355 inch ± 0.0020.
Dielectric core	Type A-1: Solid, polyethylene Diameter: 0.116 inch ± 0.004.
Outer conductor	Single braid of AWG #36, tinned copper wire. Diameter: 0.150 inch maximum. Alternate Coverage: 92.8% nominal 94.2% nominal Carriers: 12 16 Ends: 9 7 Picks/inch: 7.7 ± 10% 10.3 ± 10%
Jacket	Type IIA: PVC. Diameter: 0.195 inch ± 0.004.

ENGINEERING INFORMATION

Continuous working voltage: 1,400 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 1.240 ohms per 100 feet.

Elongation: 10 percent, minimum.

Tensile strength: Not applicable.

Engineering notes: This cable useful in general purpose low temperature applications. (See connector series “TNC”, “BNC” and “SMA” in accordance with MIL-PRF-39012, NATO preferred type NWR-2.)

REQUIREMENTS

Dimensions, configuration, and descriptions: See figure 1 and table II.

Environmental and mechanical:

Visual and mechanical examination:

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Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 5 pounds, minimum; 15 pounds, maximum.

Aging stability: +98ºC ± 2ºC.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

Cold bend: -55ºC ± 2ºC.

Dimensional stability: +85°C ± 2°C.

Inner conductor from core: 0.062 inch, maximum.

Inner conductor from jacket: 0.125 inch, maximum.

Contamination: Applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight: 0.026 pound per foot, maximum.

Electrical:

Continuity: Applicable.

Spark test: 5,000 V rms, +10%, -0%.

Voltage withstanding: 5,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 V rms, minimum.

Characteristic impedance: 50 ohms ± 2.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32.2 pF per foot, maximum.

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Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

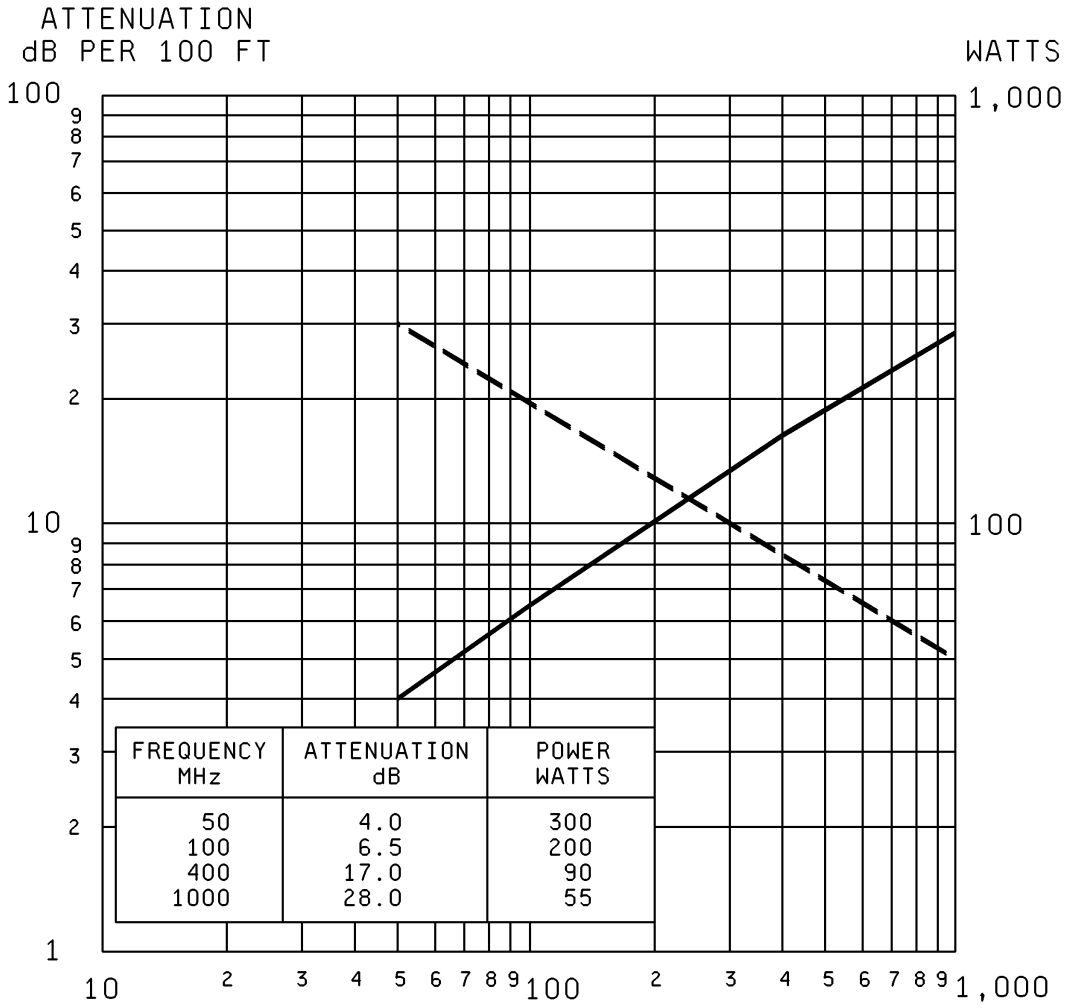
Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

PIN: M17/28-RG058. See table I.

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FREQUENCY MHz	ATTENUATION dB	POWER WATTS
50	4.0	300
100	6.5	200
400	17.0	90
1000	28.0	55

Tabulated values are for reference only.
The values on the chart represent the requirements.

FIGURE 2. Power rating and attenuation.

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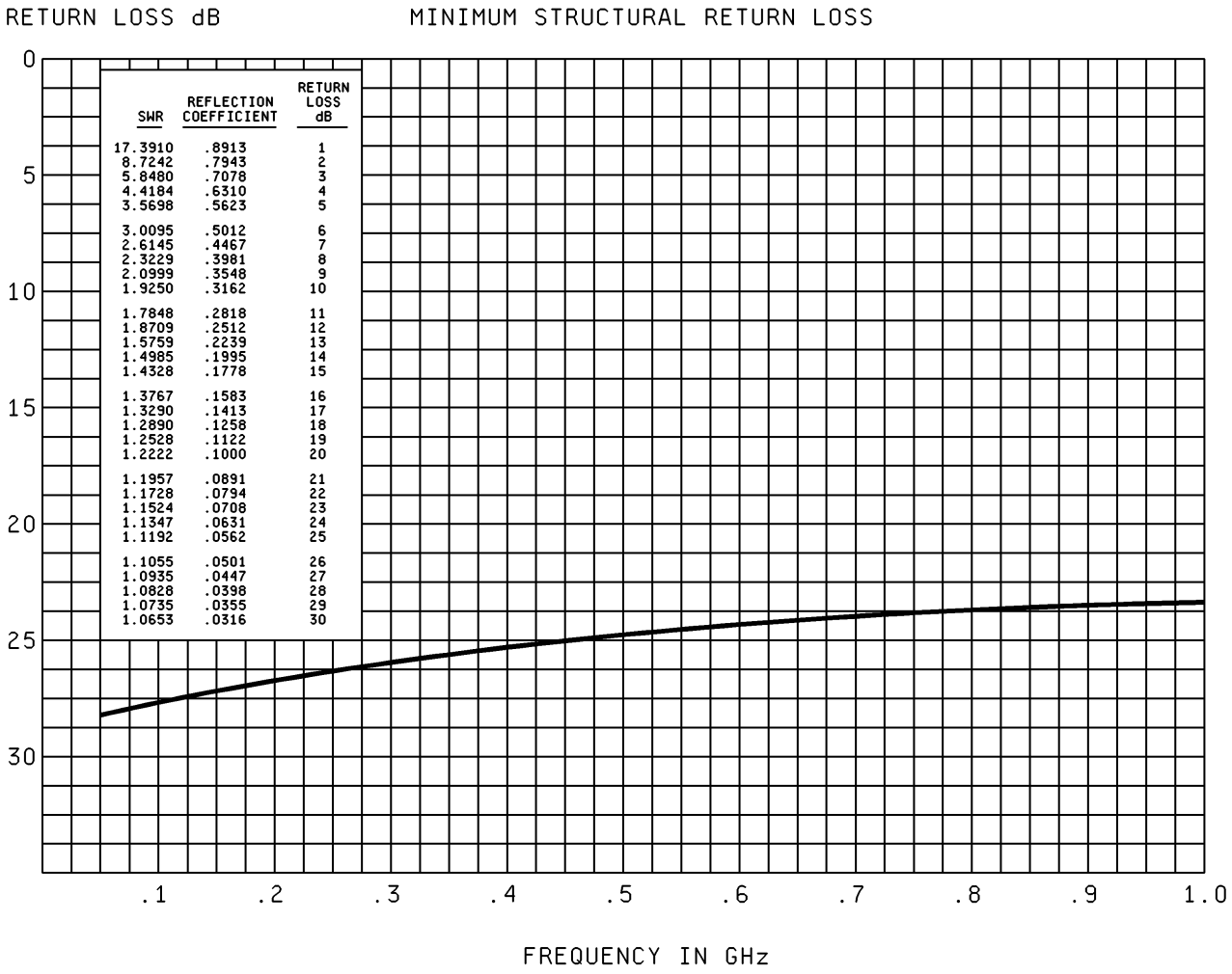


FIGURE 3. Structural return loss.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012
MIL-C-17/183

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2010-022)

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH, TD
Air Force – 19, 99
DLA - IS

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